CSE1141 Fall2019

Programming Assignment #3

Due Date: 01/12/2019 – 23:00 (No Extension)

In this assignment, you will write the following programs by using loops.

1. Write a program which finds the smallest number (excluding 1) whose sum of cubes of digits are equal to that number. The program should print the number you found as output.

   Example:
   Assume the number you are testing is 29.

   2^3 + 9^3 = 8 + 729 = 737.
   737 is not equal to 29. So the number you are searching for is not 29.

2. Write a program that prompts the user to enter *an integer number (N)* and *a character* then displays the following (2N+1)-by-(2N+1) diamond pattern like the one below:

```
Enter a number: 4

Enter a character: C

. . . . C . . . .
. . . C C C . . .
. . C C C C C . .
. C C C C C C C .
C C C C C C C C C
. C C C C C C C .
. . C C C C C . .
. . . C C C . . .
. . . . C . . . .
```

Note: Your program should execute correctly for different test cases.

Submission Instructions:

Please zip and submit all your files using filename ***YourNumberHW3.zip*** (ex: 150713852HW3.zip) to Canvas system (under Assignments tab). Your zip file should contain the followings:

1. Java source code for Q1 (Pro3_1_150713852.java)
2. Java class file for Q1 (Pro3_1_150713852.class)
3. Java source code for Q2 (Pro3_2_150713852.java)
4. Java class file for Q2 (Pro3_2_150713852.class)

Your program ***must include necessary comments with your own words to explain your actions!***

Notes:

1. Write a comment at the beginning of each program to explain the purpose of the program.
2. Write your name and student ID as a comment.
3. Include necessary comments to explain your actions.
4. Select meaningful names for your variables.
5. You are allowed to use the materials that you have learned in lectures & labs.
6. Do not use the things that you did not learn in the course.
7. **Program submissions** should be done through the Canvas class page, under the assignments tab. Do not send program submissions through e-mail. E-mail attachments will not be accepted as valid submissions.
8. You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.
9. In case of any form of **copying and cheating** on solutions, all parts will get **ZERO** grade. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties.
   All types of plagiarism will result in zero grade from the homework.
10. No late submission will be accepted

Grading:

Problem 1 (40 points)

Problem 2 (50 points)

Submission format (10 points)

- 2 java files + 2 class files
- Make sure that your class files can be executed on another computer.
- Make sure that the input/output of your program must be the same with the examples above (all informative strings & spaces).
- Comments are necessary!